

Elisabeth Kollrack

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Education

University of Vermont, Burlington, VT

May 2026

Bachelor of Science in Data Science | Minor in Mathematics | GPA: 3.98

College of Engineering and Mathematical Sciences, Patrick Leahy Honors College

- **Honors & Awards:** Summa Cum Laude, Dean's List (All Semesters)
- **Related Coursework:** Machine Learning, Software Engineering, Data Science Pinnacle, Advanced Programming, Algorithm Design, Calculus III, Advanced Statistical Methods, Linear Algebra, Database Systems
- **Technical Skills:** Python, SQL, R, Java, C++, MATLAB, JavaScript, FastAPI, Docker, Git, Snowflake, Excel

Relevant Experience

Araceli Biosciences, Tigard, OR

June 2026 – Present

Data Science Intern

- Built an LLM-assisted tool that helps scientists design fluorescence microscopy acquisition protocols and interpret instrument QC results, powered by a self-hosted language model served via vLLM behind a FastAPI backend.
- Engineered a Python pipeline to parse and summarize raw instrument QC output into structured, human-approvable reports, replacing manual review.
- Architected a multi-step web interface (React, Fluent UI, Plotly) guiding users through protocol creation, review, and refinement.
- Presented system architecture, methodology, and findings to leadership, scientists, and engineers across technical and non-technical audiences.

UVM Tutoring Center, Burlington, VT

September 2023 – May 2026

Peer Training Coordinator, Tutor

- Led weekly training sessions for 3–10 tutors, explained policies and promoted student-led learning approaches. Held office hours to support peer tutors with teaching strategies and student concerns.
- Developed a Python program to cross-reference tutor schedules with lecture times to streamline tutor availability.
- Tutored students one-on-one in calculus and computer science, providing conceptual clarity and problem-solving support.

Cox Automotive, Burlington, VT

May 2025 – August 2025

Search Engine Optimization (SEO) Intern

- Collaborated with the SEO team to analyze large-scale datasets and deliver visually engaging reports, communicating findings effectively with remote teams.
- Conducted research on Share of Voice metrics using Snowflake and Excel and built data visualizations to highlight key findings.
- Applied vector embedding techniques using Python, machine learning, and linear algebra to uncover optimization opportunities across site content and structure.

Projects

NFL Fandom Sentiment & Geographic Analysis, Undergraduate Honors Thesis

March 2025 – April 2026

- Analyzed geotagged Twitter data using Python to measure geographic dispersion of NFL fan bases and quantify team engagement.
- Created a spatial fandom radius metric using metropolitan population data and applied sentiment analysis to evaluate fan reactions to game outcomes.

Eatalyze, Machine Learning Project

May 2025

- Developed a machine learning system with a team of 3 using convolutional neural networks and XGBoost to classify food images and analyze nutrition, enabling live webcam predictions and data driven insights.

Title IX Lawsuit, Plaintiff

August 2021 – April 2022

- Filed and won a Title IX Lawsuit against Portland Public Schools to obtain equal athletic facilities for men and women (featured in ESPN documentary – [37 Words](#) (Part 4))